
Symmetries in Physics — Exercise Sheet 4

Schur's lemma and the group algebra

Variant: Questions only

Exercise 1. Let ρ be a complex irreducible representation of a finite group G on V .

- (i) Use Schur's lemma to show that every central element z acts as a scalar, $\rho(z) = \lambda(z)\mathbb{1}$.
- (ii) Suppose moreover that ρ is faithful. Show that $z \mapsto \lambda(z)$ is an injective homomorphism from the centre $Z(G)$ to $\mathbb{C} \setminus \{0\}$, and deduce that $Z(G)$ is cyclic.
- (iii) Conclude that the Klein four-group has no faithful irreducible representation. Check part (ii) on D_4 , whose centre is $\{e, r^2\}$, in the representation of Exercise 2.

Exercise 2. Take the two-dimensional representation of D_4 in Sheet 1's basis, $D(r) = R(\pi/2)$ and $D(s) = \text{diag}(1, -1)$, and the one-dimensional representation χ with $\chi(r) = -1$, $\chi(s) = 1$. Writing $D_{\mu\nu}(g)$ for the matrix elements and $n = 2$ for the dimension, verify three instances of the orthogonality of matrix elements:

$$\sum_g |D_{12}(g)|^2 = \frac{|G|}{n}, \quad \sum_g D_{11}(g) \overline{D_{22}(g)} = 0, \quad \sum_g \overline{\chi(g)} D_{11}(g) = 0.$$

Exercise 3. Find all irreducible representations of D_4 .

- (i) Show that D_4 has exactly four one-dimensional representations.
- (ii) Show that the two-dimensional representation of Exercise 2 is irreducible.
- (iii) Use $\sum_i n_i^2 = |G|$ to show that there are no others, and say how many times each one occurs in the regular representation.

Exercise 4. On $\mathbb{Z}/n\mathbb{Z}$ define the convolution and the Fourier transform:

$$(f * h)(x) = \sum_y f(y)h(x - y), \quad \widehat{f}(k) = \sum_x f(x)\omega^{-kx}, \quad \omega = e^{2\pi i/n}.$$

Prove the convolution theorem $\widehat{f * h} = \widehat{f}\widehat{h}$. Explain why it says that the basis of functions χ_k (the vectors v_k of Sheet 3, Exercise 1 — the *Fourier basis*) simultaneously diagonalises the regular representation.

Exercise 5. The quaternion group is $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ with $i^2 = j^2 = k^2 = ijk = -1$.

- (i) Find all one-dimensional representations.
- (ii) Deduce from $\sum_i n_i^2 = 8$ that exactly one two-dimensional irreducible remains, and realise it with Pauli matrices.
- (iii) Verify its irreducibility by a commutant argument. Show that only scalars commute with all the representing matrices. Explain (using Maschke's theorem) why a reducible representation would admit a non-scalar commuting operator.

Exercise 6. Without constructing any representation, use the two counting theorems of Lecture 5 to find the dimensions of the irreducible representations

- (i) of A_4 ;
- (ii) of D_6 .
- (iii) Then construct the one-dimensional representations of A_4 from the quotient $A_4/V \cong C_3$ of Lecture 2.

Exercise 7 (the coordinate functions carry the defining representation). This exercise works through Example 3.9 of Lecture 3, the functions x and y . Let D_3 act on the plane by the defining representation of Lecture 3,

$$D(r) = \frac{1}{2} \begin{pmatrix} -1 & -\sqrt{3} \\ \sqrt{3} & -1 \end{pmatrix}, \quad D(s) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix},$$

with the vertices of the triangle at $\vec{p}_1 = (0, 1)$, $\vec{p}_2 = (-\sqrt{3}/2, -1/2)$, $\vec{p}_3 = (\sqrt{3}/2, -1/2)$. Functions on the plane transform by $(\rho(g)\psi)(\vec{v}) = \psi(g^{-1}\vec{v})$.

- (i) Check that r carries $\vec{p}_1 \mapsto \vec{p}_2 \mapsto \vec{p}_3 \mapsto \vec{p}_1$. Write down r^{-1} and compute the four functions $\rho(r)x$, $\rho(r)y$, $\rho(s)x$, $\rho(s)y$ as linear combinations of the coordinate functions x and y .
- (ii) Write the matrices of $\rho(r)$ and $\rho(s)$ in the basis (x, y) , using the convention that the j -th column lists the components of the image of the j -th basis vector. Compare them with $D(r)$ and $D(s)$.
- (iii) Evaluate $\rho(r)x$ at the vertex $\vec{p}_1$. Which vertex does the value belong to, and why is that what the inverse in the definition demands?
- (iv) The $2p_x$ and $2p_y$ orbitals of an atom are $x f(|\vec{r}|)$ and $y f(|\vec{r}|)$, with one radial function f . How do they transform under D_3 ?